

RYAZHSK, Russian Federation

WMO: 278350

Lat: 53.717N

Lon: 40.117E

Elev: 127

StdP: 99.81

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.1	(c) -22.9	(d) -28.9	(e) 0.3	(f) -25.9	(g) -25.7	(h) 0.4	(i) -22.7	(j) 7.6	(k) -5.6	(l) 6.3	(m) -2.3	(n) 0.6	(o) 20	(p) 0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.5	19.9	30.3	19.6	28.3	19.0	21.5	29.0	20.7	28.2	19.9	26.8	2.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.4	18.3	13.4	23.5	17.5	12.7	22.7	62.8	28.7	60.2	28.3	57.3	27.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.9	(b) 5.2	(c) 4.4	(e) -30.8	(f) 34.0	(g) 3.8	(h) 2.6	(i) -33.5	(j) 35.9	(k) -35.7	(l) 37.4	(m) -37.8	(n) 38.8	(o) -40.6	(p) 40.7		
			DB	WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-9.1	-8.1	-2.7	6.5	15.2	17.8	20.5	19.2	12.7	6.4	0.0	-4.2
	DBStd	11.57	7.61	7.23	5.34	4.72	4.47	3.76	3.65	4.27	3.61	4.33	4.74	5.83
	HDD10.0	2512	593	506	393	128	8	1	0	0	13	128	300	442
	HDD18.3	4622	852	739	651	356	116	56	19	40	172	371	549	700
	CDD10.0	1147	0	0	0	22	171	233	325	286	94	15	1	0
	CDD18.3	216	0	0	0	0	20	38	86	68	4	0	0	0
	CDH23.3	2364	0	0	0	8	264	376	912	762	42	0	0	0
	CDH26.7	854	0	0	0	1	63	108	351	324	6	0	0	0
(14) Wind	WSAvg	1.9	2.0	2.0	2.1	2.0	1.9	1.8	1.4	1.6	1.5	1.8	2.0	2.2
(15) Precipitation	PrecAvg	543	35	27	26	34	46	56	80	54	48	49	46	42
	PrecMax	742	104	78	54	90	130	210	160	135	160	147	79	83
	PrecMin	385	5	0	6	5	1	2	11	2	3	2	19	17
	PrecStd	96	20	17	12	18	32	40	41	29	34	30	19	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.2	13.7	24.3	30.8	31.9	36.9	37.6	27.4	20.0	12.3	7.4
		MCWB	2.4	1.9	9.3	12.8	19.1	18.7	19.9	20.3	17.4	13.4	10.2	6.4
	2%	DB	2.2	2.2	8.5	20.4	28.5	29.5	32.6	32.9	24.0	17.0	9.9	5.5
		MCWB	1.4	1.2	4.7	11.6	17.5	19.1	20.1	19.9	15.7	12.7	8.3	4.9
	5%	DB	1.2	1.5	5.5	18.1	26.3	27.5	30.4	30.2	21.5	14.4	8.1	3.9
		MCWB	0.5	0.7	2.6	10.9	16.6	18.9	20.0	19.8	14.9	11.5	6.9	3.4
	10%	DB	0.4	0.7	3.6	15.5	24.2	25.2	28.1	27.3	19.4	12.5	6.2	2.1
		MCWB	-0.1	0.1	1.9	9.3	15.9	17.8	19.4	18.9	13.9	10.3	5.1	1.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.4	8.5	14.3	20.5	21.8	22.6	22.0	18.2	14.9	10.5	6.6
		MCDB	3.1	2.9	13.3	20.9	27.9	28.0	29.8	29.8	24.0	17.8	12.0	7.1
	2%	WB	1.5	1.5	5.2	12.6	18.7	20.7	21.7	21.2	16.7	13.4	8.6	4.9
		MCDB	2.0	2.0	7.9	18.5	26.5	27.2	29.0	29.9	22.7	16.0	9.7	5.3
	5%	WB	0.7	0.8	3.2	11.3	17.5	19.6	20.9	20.2	15.7	11.8	7.0	3.3
		MCDB	1.1	1.3	4.9	16.8	24.5	25.3	28.0	28.8	20.2	14.0	8.1	3.9
	10%	WB	-0.1	0.1	2.1	9.9	16.4	18.6	20.1	19.4	14.6	10.4	5.2	1.6
		MCDB	0.5	0.8	3.3	14.6	22.7	23.8	26.8	26.7	18.1	12.3	6.1	2.0
(35) Mean Daily Temperature Range	MDBR		6.0	6.8	8.0	10.3	11.9	10.6	11.5	11.8	9.9	7.3	5.0	4.7
	5% DB	MCDBR	3.7	5.0	9.8	14.8	14.9	13.6	14.7	15.4	13.9	10.1	7.6	3.8
		MCWBR	3.8	4.5	6.3	7.4	6.4	5.7	5.0	5.6	6.9	6.3	6.2	3.7
	5% WB	MCDBR	3.6	4.4	8.4	12.8	13.4	11.9	12.6	13.6	11.7	8.9	6.4	3.6
		MCWBR	3.6	4.2	5.8	7.1	6.6	5.7	5.2	5.7	6.1	6.2	5.7	3.6
(40) Clear-Sky Solar Irradiance	taub		0.296	0.336	0.357	0.428	0.393	0.388	0.419	0.439	0.385	0.361	0.331	0.299
	taud		2.109	2.024	2.068	2.089	2.306	2.343	2.272	2.201	2.332	2.324	2.229	2.124
	Ebn at Noon		635	723	804	794	854	862	824	778	780	709	595	547
	Edh at Noon		70	106	126	142	121	117	124	126	98	79	63	57
(44) All-Sky Solar Radiation	RadAvg		0.83	1.81	3.33	4.19	5.65	5.78	5.74	4.74	3.08	1.51	0.65	0.45
	RadStd		0.17	0.21	0.44	0.35	0.47	0.61	0.51	0.46	0.40	0.22	0.13	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.78	N/A	N/A	N/A	N/A	N/A	N/A	-195	+128	N/A

Nomenclature: See separate page